Project Design Phase

Problem–Solution Fit Template

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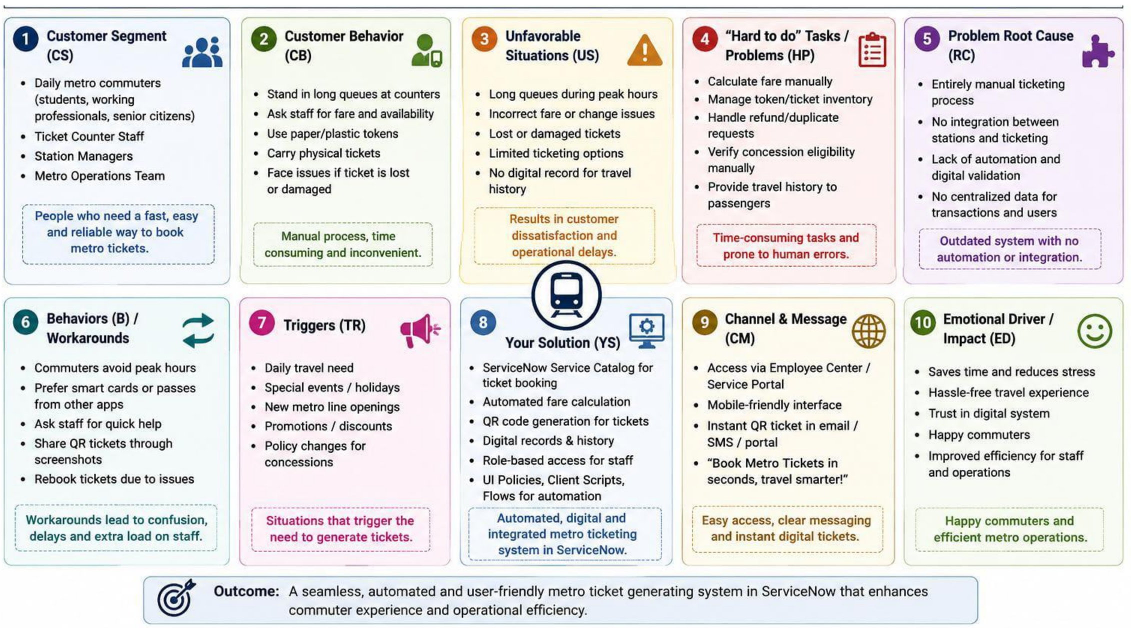
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

1. Visual Artifact: Problem–Solution Fit Canvas

Below is the structured problem–solution fit canvas summarizing the alignment between metro commuters' ticket-booking difficulties and the automated ServiceNow solution.



Theoretical Breakdown of the 10 Core Nodes

1. Customer Segment (CS)

• Theory: Identifies the primary users who experience the problem and whose needs must be addressed by the solution.

• Application: Targets metro commuters, including daily passengers, occasional travelers, students, and passengers who need quick and convenient ticket booking. The system is designed to support users with different levels of technical familiarity.

2. Customer Behavior (CB)

• Theory: Captures the current-state user journey (As-Is), including the manual or inconvenient steps users follow when the existing process does not provide an efficient digital experience.

• Application: Describes commuters waiting at ticket counters or ticket machines, selecting source and destination stations manually, calculating fares, making payments, and waiting for a physical ticket. These steps can increase waiting time during busy hours.

3. Unfavorable Situations (US)

• Theory: Defines the situations where the existing process causes the greatest inconvenience, delays, or operational impact.

• Application: Highlights long queues during peak hours, delayed ticket issuance, incorrect ticket details, difficulty in handling high passenger volumes, and inconvenience when passengers need to travel quickly.

4. "Hard to Do" Tasks / Problems (HP)

• Theory: Identifies the specific tasks that create friction, consume time, or are prone to human error.

• Application: Focuses on manually entering or selecting source and destination stations, determining the number and type of tickets, calculating the correct fare, processing payment details, and generating a valid ticket for the passenger.

5. Problem Root Cause (RC)

• Theory: Identifies the underlying process or system limitations responsible for the observed problems rather than blaming users for individual errors.

• Application: The root causes include dependence on manual ticket-booking processes, limited automation, separate handling of station and fare information, manual validation, and lack of an integrated digital workflow for ticket generation and QR-code delivery.

6. Behaviors (B) / Workarounds

• Theory: Examines the alternative actions users take when the normal process is slow, inconvenient, or unavailable.

• Application: Commuters may wait in different queues, seek assistance from station staff, repeatedly verify fare information, or postpone booking during peak periods.

Station staff may also need to manually correct booking details or handle repeated passenger queries.

7. Triggers (TR)

• Theory: Identifies the events or circumstances that cause users to interact with the system and create an immediate need for the service.

• Application: Common triggers include the need to travel by metro, peak-hour travel, last-minute journeys, increased passenger volume, and the need for fast digital ticket generation before entering the metro station.

8. Your Solution (YS)

• Theory: Represents the target-state (To-Be) solution and explains how technology addresses the identified root causes through automation, validation, and structured workflows.

• Application: The Metro Ticket Generating System in ServiceNow uses a Service Catalog item to collect source station, destination station, passenger type, number of tickets, ticket type, and payment details. Client Scripts and UI Policies support validation and dynamic form behavior. Fare calculation is automated using station details and journey information, while the workflow can generate a digital ticket with a QR code and ticket expiry information. ServiceNow Flow Designer can automate the ticket-processing workflow.

9. Channel & Message (CM)

• Theory: Defines where users access the solution and the clear value proposition communicated to them.

• Application: The solution can be provided through the ServiceNow Service Catalog / Employee Center interface. The core message is simple: book a metro ticket digitally, calculate the fare automatically, complete payment, and receive a QR-based ticket quickly without waiting in a long queue.

10. Emotional Driver / Impact (ED)

• Theory: Describes the positive emotional and practical outcome created when the solution removes the user's major pain points.

• Application: The system changes the commuter experience from waiting, confusion, and uncertainty to convenience, speed, confidence, and predictability. It also supports metro staff by reducing repetitive manual work and improving the consistency of ticket-processing operations.

Problem–Solution Alignment Summary

The main problem is the time-consuming and manually dependent metro ticket-booking process. The proposed Metro Ticket Generating System in ServiceNow addresses this problem through a centralized Service Catalog interface, automated fare calculation, validation, workflow automation, digital payment handling, and QR-code-based ticket generation. The solution is intended to reduce waiting time, minimize manual errors, improve commuter convenience, and provide a structured digital ticketing experience.